EE115 Analog Circuits
模拟电路

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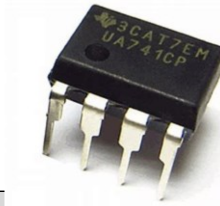
Outline

- Non-ideal Opamp
- SEDTRA/SMITH book page 96-100；105-115

Review: Ideal Op-Amp

- Infinite input impedance
  - No current goes in, virtual open circuit (虚断)
- Zero output impedance
- Infinite open-loop gain, A
  - Virtual short circuit between the inverting and non-inverting input ports (虚短)
- Infinite bandwidth
- Infinite common-mode rejection
  - Common mode rejection ratio (CMRR)

Electrical Characteristics of a Real Op-Amp



7.3 Electrical Characteristics μA741C, μA741M

at specified virtual junction temperature, $V_{CC\pm} = \pm 15$ V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	T_A (1)	μA741C			μA741M			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
V_{IO}	Input offset voltage	$V_O = 0$	25°C		1	6		1	5	mV
			Full range			7.5		±15	6	
$\Delta V_{IO(adj)}$	Offset voltage adjust range	$V_O = 0$	25°C		±15			20	200	mV
I_{IO}	Input offset current	$V_O = 0$	25°C		20	200			500	nA
			Full range			300			500	
I_{IB}	Input bias current	$V_O = 0$	25°C		80	500		80	500	nA
			Full range			800			1500	
V_{ICR}	Common-mode input voltage range		25°C	±12	±13		±12	±13		V
			Full range	±12			±12			
V_{OM}	Maximum peak output voltage swing	$R_L = 10$ kΩ	25°C	±12	±14		±12	±14		V
		$R_L \geq 10$ kΩ	Full range	±12			±12			
		$R_L = 2$ kΩ	25°C	±10			±10	±13		
		$R_L \geq 2$kΩ	Full range	±10			±10			
A_{VD}	Large-signal differential voltage amplification	$R_L \geq 2$kΩ	25°C	20	200		50	200		V/mV
		$V_O = \pm 10$ V	Full range	15			25			
r_i	Input resistance		25°C	0.3	2		0.3	2		MΩ
r_o	Output resistance	$V_O = 0$, See (2)	25°C		75			75		Ω
C_i	Input capacitance		25°C		1.4			1.4		pF
CMRR	Common-mode rejection ratio	$V_{IC} = V_{ICRmin}$	25°C	70	90		70	90		dB
			Full range	70			70			
k_{SVS}	Supply voltage sensitivity ($\Delta V_{IO}/\Delta V_{CC}$)	$V_{CC} = \pm 9$ V to ±15 V	25°C		30	150		30	150	μV/V
			Full range			150			150	
I_{OS}	Short-circuit output current		25°C		±25	±40		±25	±40	mA
I_{CC}	Supply current	$V_O = 0$, No load	25°C		1.7	2.8		1.7	2.8	mA
			Full range			3.3			3.3	
P_D	Total power dissipation	$V_O = 0$, No load	25°C		50	85		50	85	mW
			Full range			100			100	

Switching Characteristics of a Real Op-Amp

over operating free-air temperature range, $V_{CC\pm} = \pm 15$ V, $T_A = 25°C$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS	μA741C			μA741M			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
t_r	Rise time	$V_I = 20$ mV, $R_L = 2$ kΩ, $C_L = 100$ pF, See Figure 1		0.3			0.3		μs
	Overshoot factor			5%			5%		—
SR	Slew rate at unity gain	$V_I = 10$ V, $R_L = 2$ kΩ, $C_L = 100$ pF, See Figure 1		0.5			0.5		V/μs

Imperfections in Practical Op-Amps

- Finite open-loop gain ($A_o < \infty$)
- Finite input resistance ($R_i < \infty$)
- Non-zero output resistance ($R_o > 0$)
- Finite bandwidth
- Offset (偏移) voltage
- Input bias and offset currents

Offset (偏移) Voltage

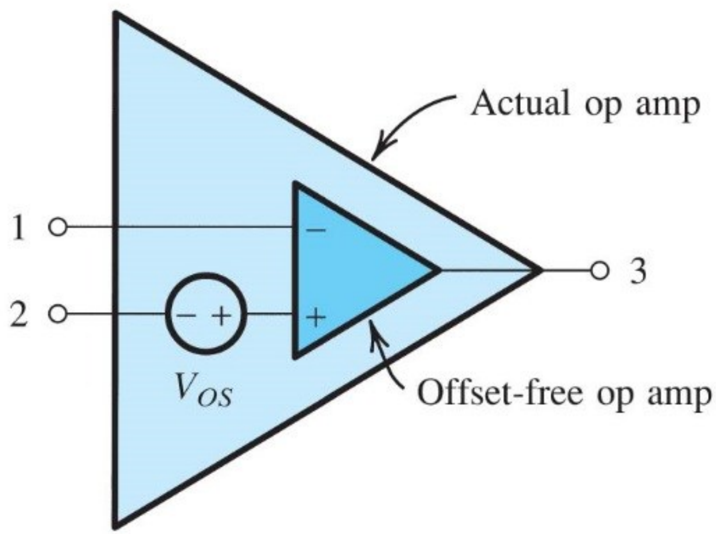


Figure 2.28 Circuit model for an op amp with input offset voltage V_{OS}.

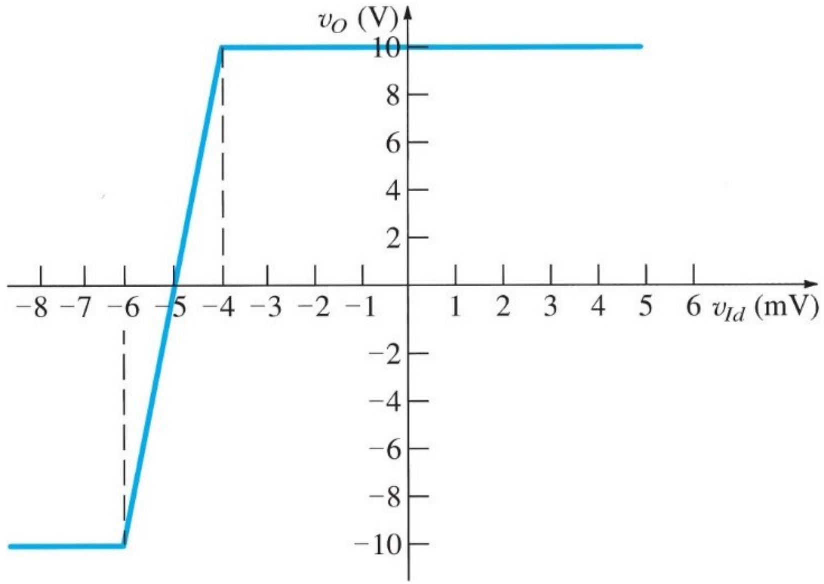
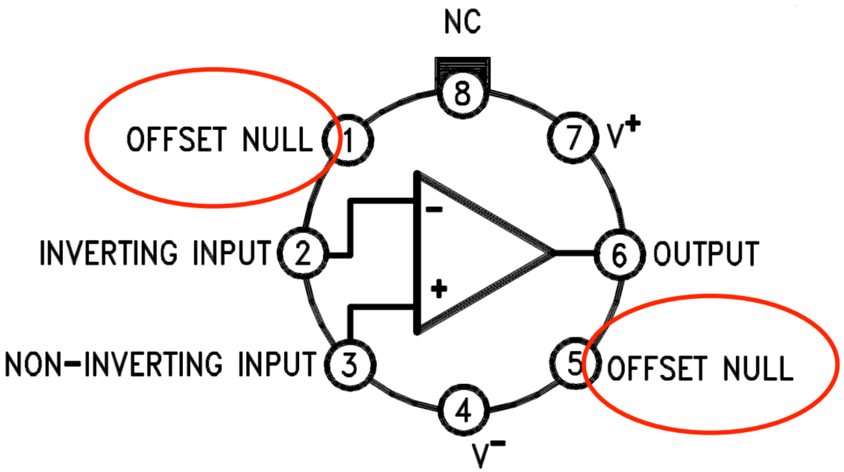


Figure E2.21 Transfer characteristic of an op amp with V_{OS} = 5 mV.

Trimming of Offset Voltage



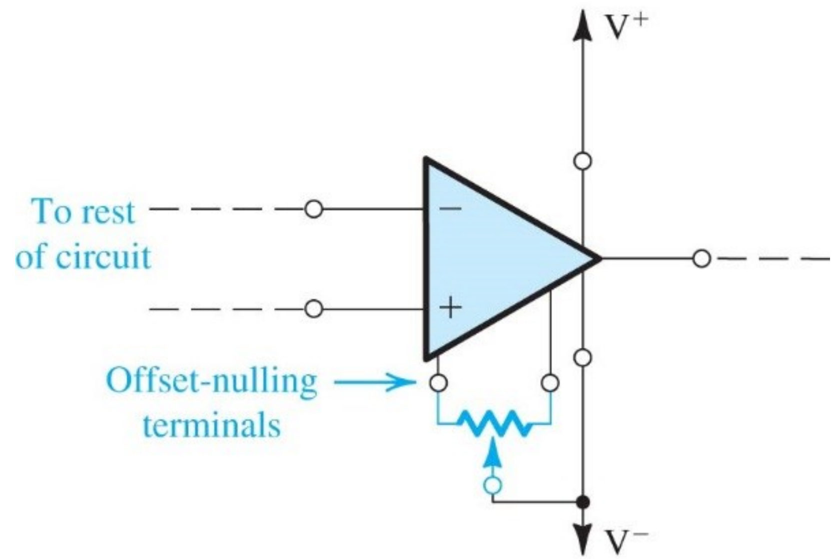
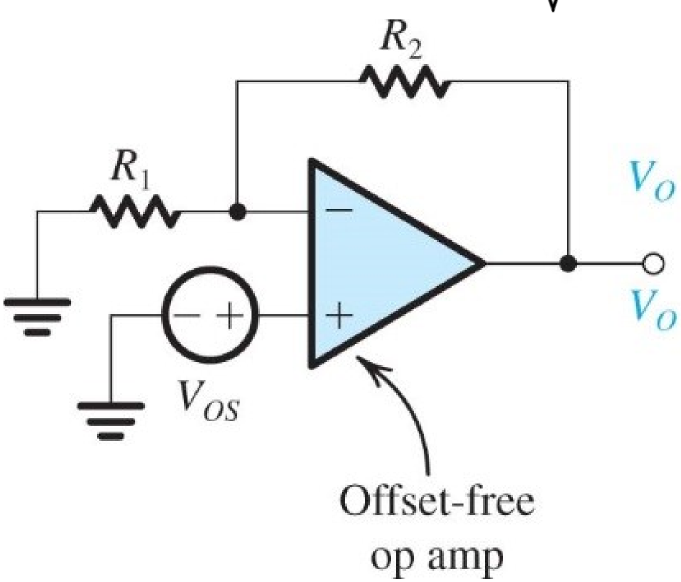


Figure 2.30 The output dc offset voltage of an op amp can be trimmed to zero by connecting a potentiometer to the two offset-nulling terminals. The wiper of the potentiometer is connected to the negative supply of the op amp.



$$V_O = V_{OS}\left(1 + \frac{R_2}{R_1}\right)$$

Figure 2.29 Evaluating the output dc offset voltage due to V_{OS} in a closed-loop amplifier.

Frequency Response of Open-Loop Op Amp

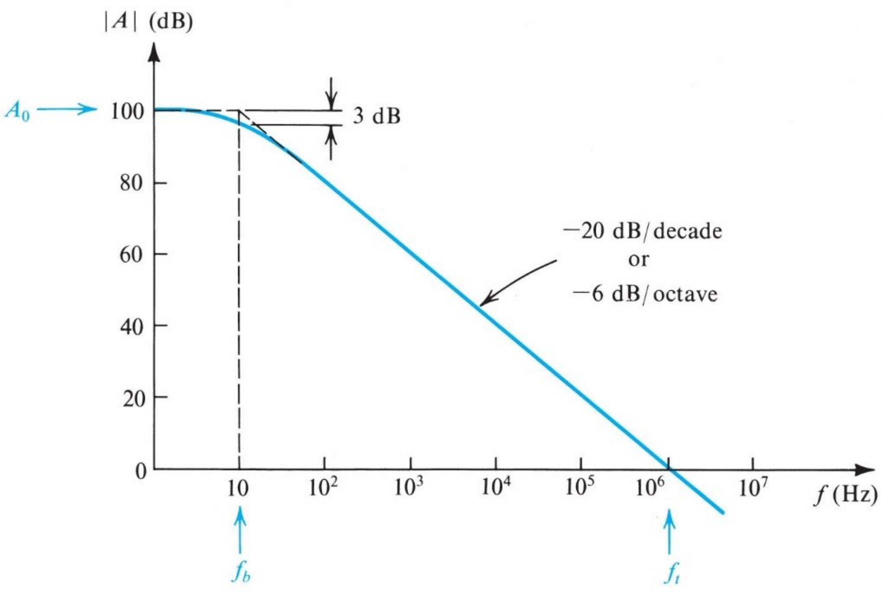


Figure 2.39 Open-loop gain of a typical general-purpose, internally compensated op amp.

- *Open-loop transfer function:*

$$A(j\omega) = \frac{A_0}{1 + j\omega/\omega_b}$$

- *At higher frequency:*

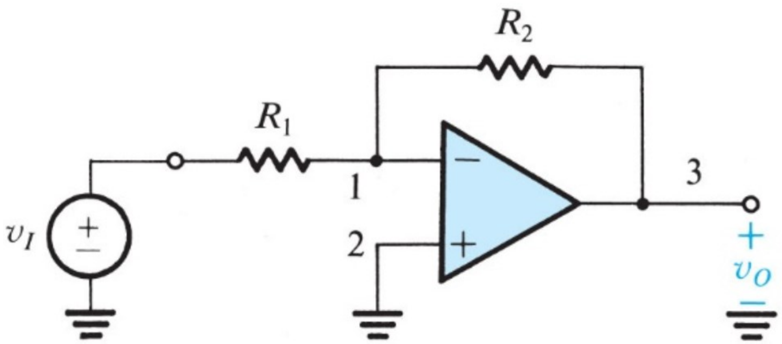
$$A(j\omega) \simeq \frac{A_0\omega_b}{j\omega} \qquad |A(j\omega)| = \frac{A_0\omega_b}{\omega}$$

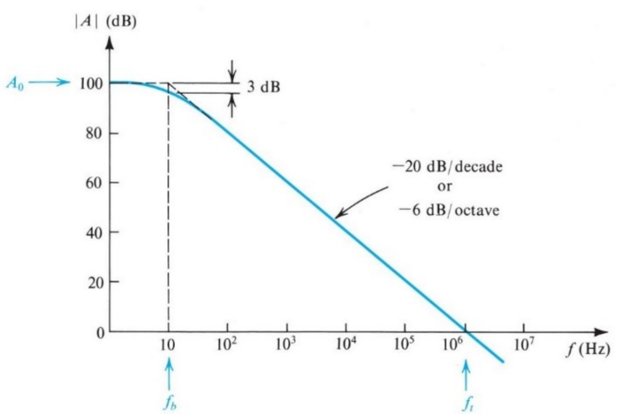
- *Unity-gain bandwidth*

$$\omega_t = A_0\omega_b$$

Single pole response with a dominant pole at ω_b

Frequency Response of Closed-Loop Op Amp (Inverting Amplifier Example)





- *Close-loop transfer function:*

$$\frac{V_o}{V_i} = \frac{-R_2/R_1}{1+(1+R_2/R_1)/A}$$

- *Include the bandwidth-limited A:*

$$\frac{V_o(s)}{V_i(s)} = \frac{-R_2/R_1}{1+\frac{1}{A_0}\left(1+\frac{R_2}{R_1}\right)+\frac{s}{\omega_t/(1+R_2/R_1)}}$$

- *If:* $A_0 \gg 1+R_2/R_1$

$$\frac{V_o(s)}{V_i(s)} \simeq \frac{-R_2/R_1}{1+\frac{s}{\omega_t/(1+R_2/R_1)}}$$

- *Close-loop corner frequency:*

$$\omega_{3\text{dB}} = \frac{\omega_t}{1+R_2/R_1}$$

- *GBW unchanged*

Frequency Response of Closed-Loop (Noninverting Amplifier Example)

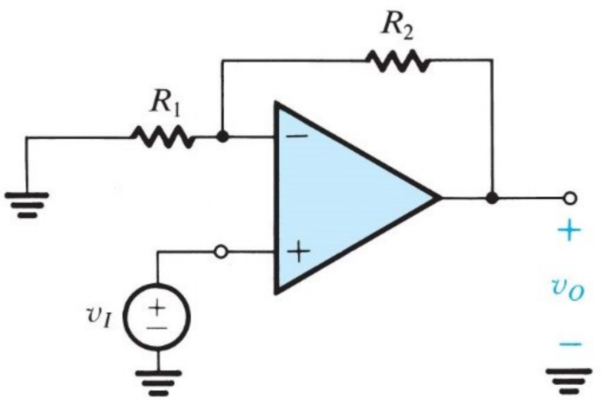
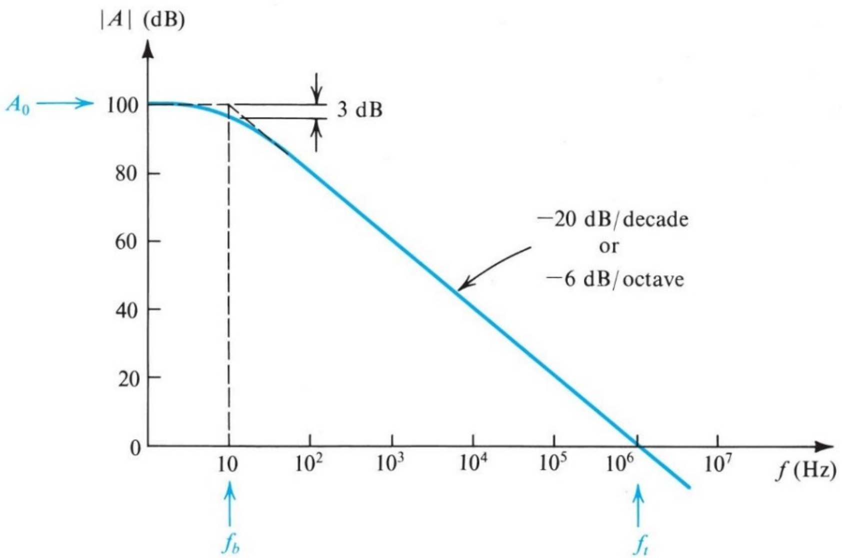


Figure 2.12 The noninverting configuration.



- *Close-loop transfer function:*

$$\frac{V_o}{V_i} = \frac{1 + R_2/R_1}{1 + (1 + R_2/R_1)/A}$$

- *Include the bandwidth-limited A:*

$$\frac{V_o(s)}{V_i(s)} \simeq \frac{1 + R_2/R_1}{1 + \dfrac{s}{\omega_t/(1 + R_2/R_1)}}$$

Close-loop corner frequency:

$$\omega_{3\text{dB}} = \frac{\omega_t}{1 + R_2/R_1}$$

GBW unchanged

Output Saturation

- The output voltage swing is limited by
  - 1. Saturation voltage (usually a volt or two lower than power supply voltage)
  - 2. Maximum output current (in case of small load resistance)
- Output waveform appears to be “clipped” when either condition happens

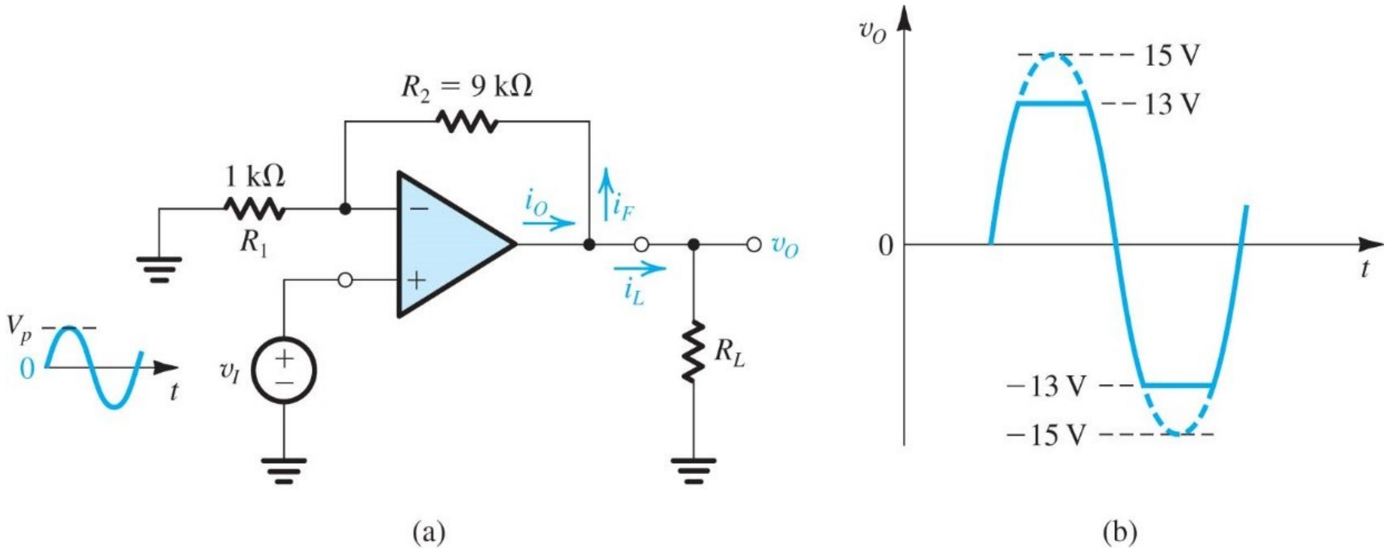


Figure 2.42 (a) A noninverting amplifier with a nominal gain of 10 V/V designed using an op amp that saturates at ±13-V output voltage and has ±20-mA output current limits. **(b)** When the input sine wave has a peak of 1.5 V, the output is clipped off at ±13 V.

Slew Rate (压摆率)

- Amplifier output is limited by "slew rate": maximum rate of change possible at output

$$SR = \left.\frac{dv_o}{dt}\right|_{max}$$

- SR is specified in datasheet in V/μs.

- Note: SR limit is different from bandwidth limit:
  - Limited bandwidth is a linear phenomenon, it does not result in changing the shape of an input sinusoid
  - SR limitation can cause nonlinear distortion to input sinusoidal signal.

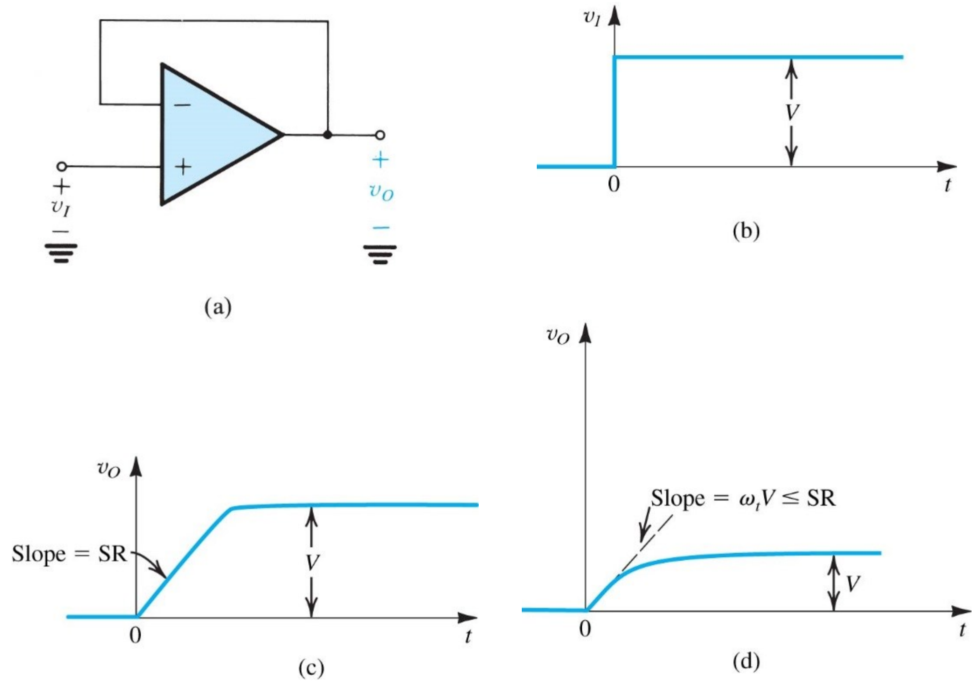
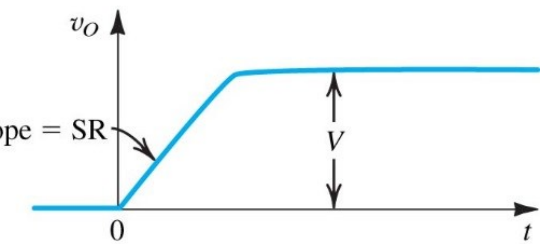


Figure 2.43 (a) Unity-gain follower. **(b)** Input step waveform. **(c)** Linearly rising output waveform obtained when the amplifier is slew-rate limited. **(d)** Exponentially rising output waveform obtained when V is sufficiently small so that the initial slope ($\omega_t V$) is smaller than or equal to SR.

Comparison of Slew Rate and Bandwidth Limits

For step function input waveform, both SR and bandwidth limits cause the output to rise with a finite slope, but there is an important difference:

Slew rate limited output:
Slope = SR at all times



(c)

- *Transfer function:*

$$\frac{V_o}{V_i} = \frac{1}{1 + s/\omega_t}$$

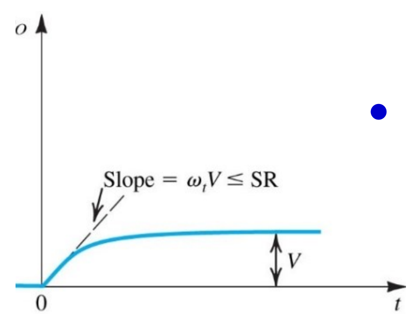
Bandwidth limited output:
Slope = $\omega_t V^* exp(-\omega_t t)$ < SR
(V is the steady state output voltage)

In response to an input step signal of height S, a low-pass STC circuit (with a dc gain $K = 1$) produces the waveform shown in Fig. E.10. Note that while the input rises from 0 to S at $t = 0$, the output does not respond immediately to this transient and simply begins to rise exponentially toward the *final* dc value of the input, S. In the long term—that is, for $t \gg \tau$ —the output approaches the dc value S, a manifestation of the fact that low-pass circuits faithfully pass dc.

The equation of the output waveform can be obtained from the expression

$$y(t) = Y_\infty - (Y_\infty - Y_{0+})e^{-t/\tau} \quad \text{(E.9)}$$

where Y_∞ denotes the *final* value or the value toward which the output is heading and Y_{0+} denotes the value of the output immediately after $t = 0$. This equation states that *the output at*



(d)

- *Time domain:*

$$v_O(t) = V(1 - e^{-\omega_t t})$$

Full-Power Bandwidth

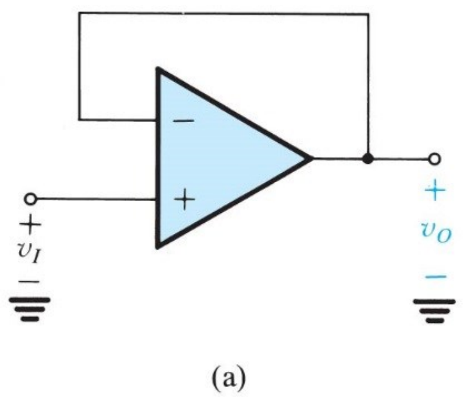


Figure 2.43 (a) Unity-gain follower.

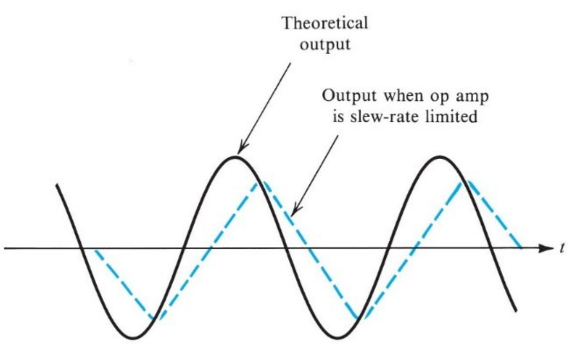


Figure 2.44 Effect of slew-rate limiting on output sinusoidal waveforms.

- For sinusoidal input to unity-gain follower:

$$v_I = V_i \sin \omega t$$

- Rate of change:

$$\frac{dv_I}{dt} = V_i \omega \cos \omega t \leq SR$$

- **Full-power bandwidth**:
  The frequency at which SR-limited distortion starts to occur for an output sinusoid with maximum rated output voltage, V_{omax},

$$\omega_M V_{omax} = SR$$

$$f_M = \frac{SR}{2\pi V_{omax}}$$

Summary

- Open loop gain
- Input impedance
- Output impedance
- Input offset & Bias
- CMRR
- Saturation
- Unity-gain bandwidth
- Slew rate
- Full-power bandwidth

Homeworkx-due March xx 11p.m.

SEDRA/SMITH Book chapter xx problem xxx